Predicting 30-Day Hospital Readmissions Using the UCI Diabetes Dataset

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ISYE 6704 Data Mining & Statistical Learning

Abstract

Hospital readmissions within 30 days present significant implications for patient and system costs. This analysis aims to identify key factors associated with readmission risk and to evaluate predictive modeling approaches. The dataset included over 50 features across hospital utilization, clinical diagnoses, and demographic factors. Categorical encoding, feature engineering, and statistical feature selection were used to reduce the dataset's dimensionality.

A Logistic Regression model for interpretability and a Random Forest model for predictive performance were implemented. Both models began with independent feature selection to identify significant predictors. Because readmission in <30 days was the minority class, the F1 score and the area under the receiver operating characteristic (ROC) curve (AUC) were used due to the high class imbalance in this dataset.

Results included moderate predictive accuracy for logistic regression and high predictive accuracy for Random Forests. Although the ensemble method achieved higher predictive accuracy, both models struggled to accurately predict the minority class, readmission <30 days, even after feature selection, as they tended to overpredict the majority class.

While predictive performance has been difficult to achieve, the significant features selected across both models can still serve as a reference point for providers as factors that impact readmission rates. Key significant features were centered around prior hospital utilization metrics, comorbidity score, discharge type, and admission type.

Introduction

Given the financial burden on patients, the emotional distress associated with readmission, and the strain on healthcare resources, the 30-day readmission rate has become a central metric in the healthcare system. Substantial policy changes over the last few decades have been made to disincentivize hospitals from having high readmission rates. Most notably in 2012, the Centers for Medicare & Medicaid Services (CMS) introduced the Hospital Readmissions Reduction Program (HRRP), which imposes financial penalties on hospitals with higher-than-expected readmission rates (Figueroa et al., 2024). With inherent costs of readmission for patients and hospitals, compounded by policy-related penalties, predicting patient readmission rates has become a critical priority for improving care outcomes and reducing healthcare costs.

Project Data Source

This project will utilize the Diabetes 130-US Hospitals for years 1999-2008 dataset provided by the UC Irvine Machine Learning Repository.

Data Source Link:
https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008

Based on the source documentation, the data provide 10 years of historical patient encounter data across 130 US hospitals, filtered to include only encounters in which the patient was diagnosed with diabetes, underwent lab work, was administered diabetic medications, and stayed between 1 and 14 days.

Problem Statement

The motivation of this project is to analyze the inpatient dataset to identify key predictors of 30-day hospital readmissions among patients with diabetes. Key questions explored include:

1. Which patient and hospital-related factors are most strongly associated with 30-day readmissions among patients with diabetes?
2. How does comorbidity burden, measured using the Charlson Comorbidity Index (CCI), influence the likelihood of 30-day readmission?
3. To what extent do prior healthcare utilization variables, like inpatient visits, emergency visits, and length of stay, predict readmission risk?
4. How do Logistic Regression and Random Forest models compare in their ability to predict 30-day readmissions, particularly under conditions of class imbalance?

This data set poses a few challenges that make prediction with traditional machine learning techniques difficult. For one, the target variable *readmitted<30* is highly imbalanced. Only 8.8% of patient encounters were characterized as readmissions within 30 days. Secondly, this is a high-dimensional dataset with almost 50 features. This introduces the risk of overfitting as the number of variables approaches the number of data points. The risk of overfitting in this data set is exacerbated by the fact that many features are categorical. If categorical variables are not hierarchical, they will require one-hot encoding, by which each category becomes its own column of binary values.

Proposed Methodology

Two modeling approaches were leveraged in this analysis. First, a logistic regression model was used as a baseline due to its interpretability. Hypothesis testing was used to identify statistically significant variables using Wald test statistics. High penalties via class weighting on incorrect classifications in the minority class were used to address the class imbalance.

Second, a Random Forest model was implemented. The Random Forest algorithm is an ensemble method that generally performs better than any standalone model, as each tree is built using a bootstrap sample of the training data, and at each split, only a random subset of variables is considered. This approach reduces correlation among trees and is intended to lower overall prediction error. (Mei, n.d.). Hyperparameter tuning was used to identify the best parameters. Conveniently, Random Forest estimates variable importance from out-of-bag observations, serving as an internal cross-validation mechanism that eliminates the need for explicit variable selection (Mei, n.d.). The top predictive features were those that decreased node impurity across many trees.

Because a poorly learned model can still achieve high accuracy by simply predicting the class that holds the majority, in this case not readmitted in less than 30 days, model performance was assessed using sensitivity-specificity and precision-recall metrics, specifically F1 score and the area under the receiver operating characteristic (ROC) curve (AUC) (Brownlee, 2021). The ROC curve assesses classification performance across all possible threshold values by plotting true positive rate against false positive rate, providing a threshold-independent measure of model performance.

The AUC performance of both models was compared using k-fold cross-validation.

All analyses were conducted in Python using packages: NumPy, pandas, Matplotlib, Seaborn, scikit-learn, statsmodels.

Data Overview and EDA

Preliminary Data Cleaning

In total, the data set has 101,766 patient encounter records with 50 features, including:

- Demographic (race, age, gender)
- Insurance providers
- Admission types
- Discharge dispositions
- Medications administered
- Medication changes
- Lab results
- Clinical diagnoses

Columns were audited for variance and null rate. Variables with a high null rate included: weight, Max Glucose Serum result, A1C result, provider medical specialty, and payer code. Because this analysis focused on comorbidity factors based on diagnosis codes and healthcare utilization variables, these features were dropped. At least half of the features were dedicated to different medications, including whether they were prescribed and whether they were changed during a patient's visit. Given the existing variables “num_medications” and “change” (true if the patient had a medication change at a documented visit), all medication features were also removed. Because the admission source provided similar information to the admission type, the admission source was also dropped as a feature.

Exploratory Data Analysis (EDA)

The target variable ‘readmitted’ included “<30” for patients readmitted within 30 days of discharge, “>30” if the patient was admitted in greater than 30 days, and “NO” for those not readmitted. Upon

visualizing the distribution of the target variable, it was evident that readmission was highly imbalanced. Only 8.8% of patient encounters were characterized as readmissions within 30 days.

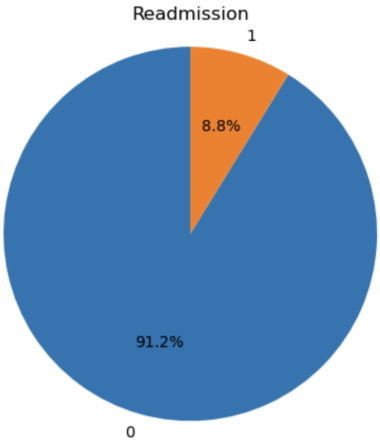

Figure 1. *Class Imbalance of Target Variable "readmitted_<30"*

The first part of the EDA included an assessment of categorical variables to see how they vary across readmission status (see appendix). Age, diagnosis category, and discharge disposition showed the strongest signals for readmission risk. Key findings include:

- Approximately 77% of patients were Caucasian, and 54% were female, indicating little variation in demographic variables.
- About 45% of readmitted patients were aged 70-90, suggesting a higher readmission risk among older patients.
- The most common diagnosis categories among readmitted patients were Circulatory, Endocrine, Respiratory, and Genitourinary conditions.
- 80% of readmitted patients were prescribed a diabetes medication, compared to 77% of non-readmitted patients. This could indicate slightly higher treatment intensity.
- A slightly higher proportion (+3%) of readmitted patients experienced a change in medication during their encounter.
- Readmitted patients were more often discharged to supportive care facilities, while non-readmitted patients were more often discharged home.
- The majority of encounters (over 50%) were emergency admissions, with readmitted patients having a slightly higher proportion (+2%) of emergency admissions compared to those not readmitted within 30 days.

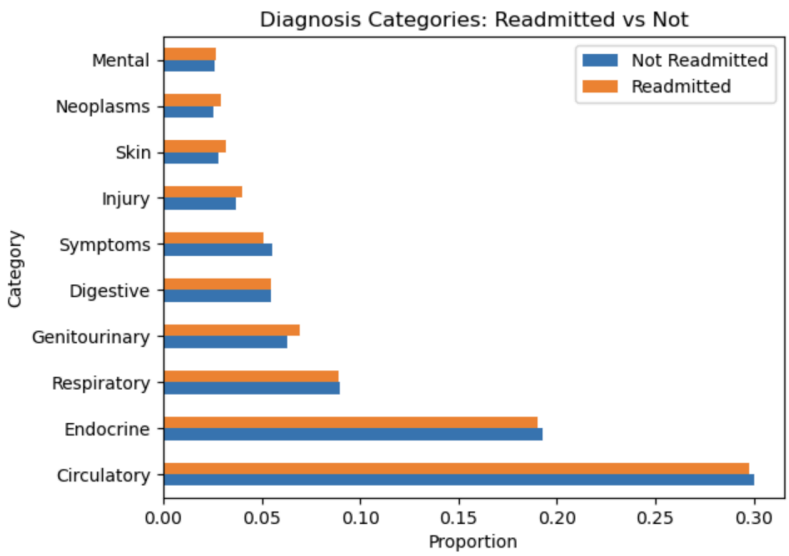

Figure 2. *Diagnosis Category Frequency for Readmitted and Not Readmitted*

The EDA of diagnosis categories showed some variation from first-level to third-level diagnoses, but overall, diagnosis categories were similar across readmitted and non-readmitted patients. To supplement the context provided by diagnosis categories, a Charlson Comorbidity Index (CCI) score was constructed using ICD-9 diagnosis codes to quantify patients' comorbidity burden (O'Connell & Lim, 2000). Using an existing lookup of conditions and their associated weights (Dewit, 2023) , the resulting scores were summed at the patient level to create an overall comorbidity measure used in subsequent analyses.

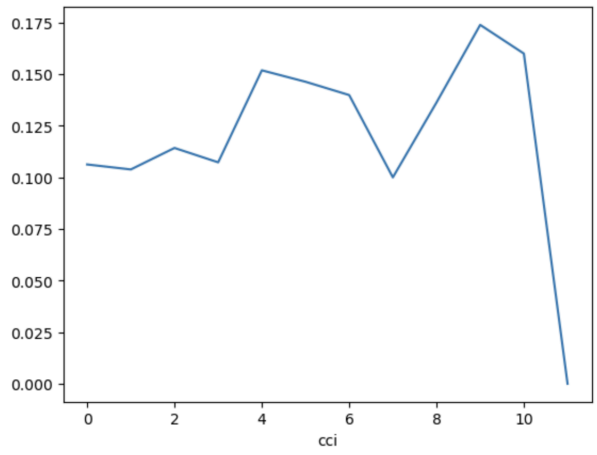

Figure 3. *Mean Readmission by Charlson Comorbidity Index (CCI)*

Figure 3 captures CCI or disease burden and its association with mean readmission. From this, it appears that CCI is positively associated with readmission at levels below 8. At higher CCI levels,

rates of death or hospice discharge are much higher (14-50%), which may explain the observed decline in readmission among the most medically burdened patients (see appendix).

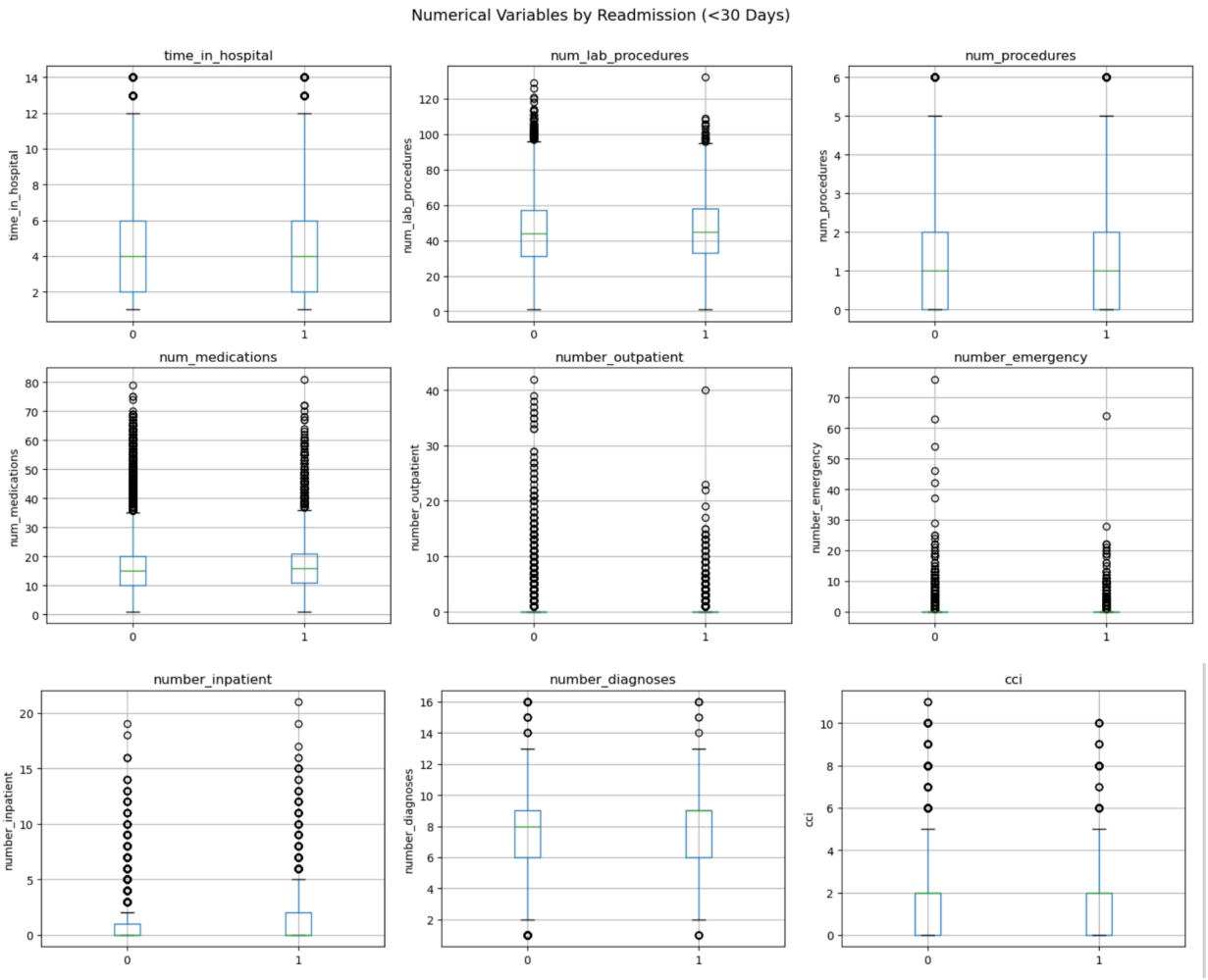

Figure 4. *Numeric Feature Distributions by Readmission Status*

across most variables, readmitted patients exhibit higher medians and a wider spread, with more extreme outliers. Boxplots of numerical variables showed that readmitted patients tend to have higher healthcare utilization, but also higher variability across these features.

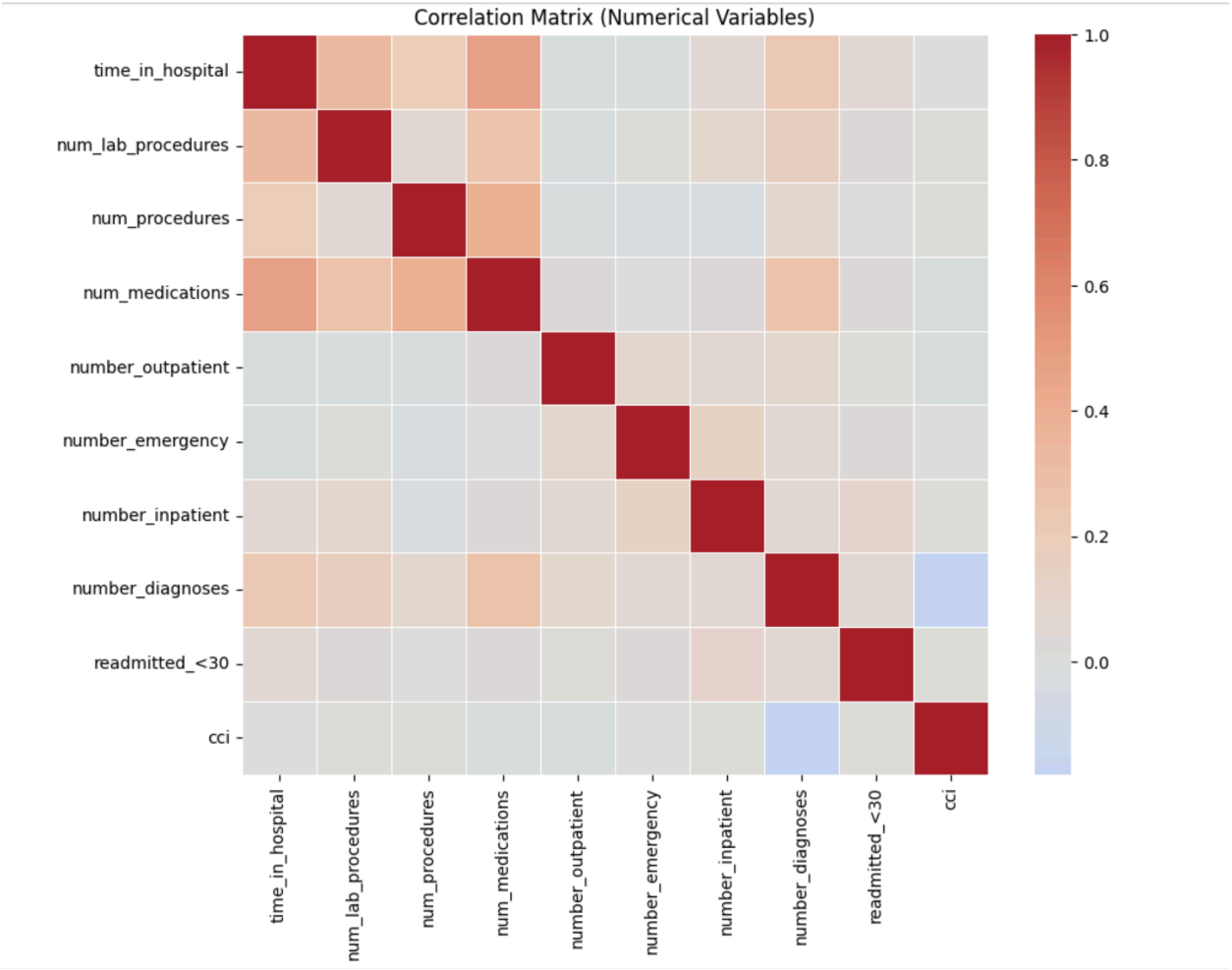

Figure 5. *Correlation Matrix for Numerical Features*

The correlation matrix revealed moderate relationships among hospital utilization measures like time in hospital, number of medications, and number of lab procedures, but overall, correlations between individual numerical variables and readmission were weak (~0.1-0.5). This suggests that readmission risk is driven by potentially non-linear interactions rather than one single predictor.

Analysis & Results

Logistic Regression

As a baseline approach, a logistic regression model was implemented to model the probability of 30-day readmission as a function of patient and hospital characteristics. Following the hypothesis testing approach to variable selection discussed in lecture, model selection was conducted by testing whether subsets of regression coefficients were equal to zero (Mei, n.d.). An initial full model

was fitted using MLE. Then, using Wald test statistics, predictors with p-values < 0.05 were retained to construct a reduced model. The reduced model used the following eleven features:

Variable	Coefficient	Odds Ratio
Intercept	-3.082	0.046
time_in_hospital	0.019	1.019
num_lab_procedures	0.002	1.002
number_emergency	0.077	1.080
number_inpatient	0.347	1.415
number_diagnoses	0.039	1.040
cci	0.072	1.074
discharge_group_Home	-0.457	0.633
discharge_group_Long_Term_Care	-1.441	0.237
discharge_group_Psych	1.567	4.792
discharge_group_Rehab	1.065	2.901
age_num	0.005	1.005

Figure 6. *Logistic Regression Summary Results for Selected Features and Odds Ratio*

The most influential predictors of readmission were related to prior healthcare utilization and discharge disposition. More specifically, the number of inpatient visits increased the odds of readmission (OR = 1.415), indicating that patients with a high number of hospitalizations are at substantially higher risk of readmission. Discharge disposition also showed an increase in odds of readmission with patients discharged to psych (OR = 4.792) and rehabilitation facilities (OR = 2.901). On the other hand, patients discharged home (OR = 0.633) or to long-term care (OR = 0.237) had lower odds. The effects of these features may suggest that differences in care pathways and patient severity can be key determinants of readmission. Comorbidity burden (CCI) and prior emergency visits were also positively associated with readmission, though with more moderate effect sizes.

To evaluate predictive performance, a parallel logistic regression model was developed using the scikit-learn library. The reduced set of statistically significant predictors identified through the variable selection process was used. The dataset was partitioned into training and test sets using an 80/20 stratified split to preserve class proportions.

	Predicted No	Predicted Yes
Actual No	8636	4104
Actual Yes	644	611

Figure 7. *Logistic Regression Confusion Matrix*

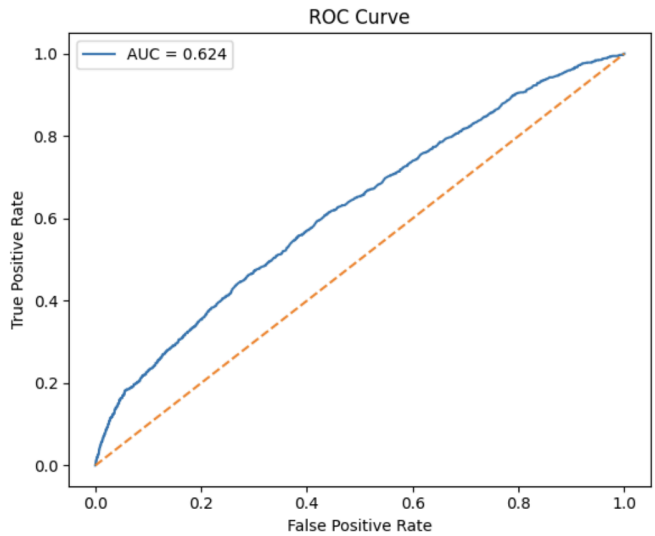

Figure 8. *Logistic Regression ROC and AUC*

According to the confusion matrix in **Figure 8**, 611 of 1255 (49%) readmitted cases were correctly predicted by the reduced logistic regression model. With an F1 score of 0.212 and an AUC of 0.624, this logistic regression model has poor predictive performance with limited ability to accurately identify positive readmission cases despite capturing significant features.

Random Forest

First, a default Random Forest model without tuning was fit to the training data, yielding an F1 score of 0.008 and an AUC of 0.6. Using hyperparameter tuning with GridSearchCV to optimize cross-validated ROC, the best model parameters were 400 trees, log2 feature selection at each split, and a minimum of 5 samples per leaf. Interestingly, the optimized model had no class weighting applied, suggesting that models that showed a preference for the minority class didn't actually improve performance. The confusion matrix for the tuned model shows the poor sensitivity (true positive rate) of the tuned random forest. While all 12,740 non-readmitted patients were classified correctly, zero out of the 1,255 readmitted patients were successfully classified.

	Predicted No	Predicted Yes
Actual No	12740	0
Actual Yes	1255	0

Figure 9. *Random Forest Confusion Matrix*

With an AUC of 0.637, the model performed better than a random guess in distinguishing between readmitted and non-readmitted patients.

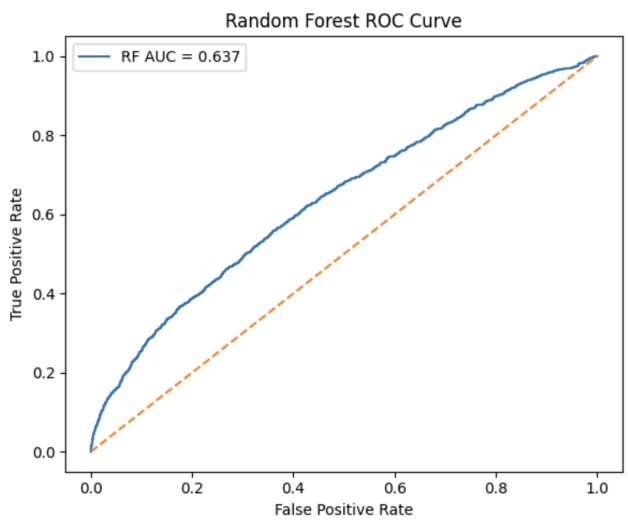

Figure 10. *Random Forest ROC Curve*

The feature importance results provide additional insight into which variables it identified as significant. The most important variables identified, specifically the number of lab procedures, the number of medications, time in hospital,number of procedures, diagnoses, inpatient visits, discharge (rehab or home), and age, are those that contribute most to reducing classification error across the ensemble.

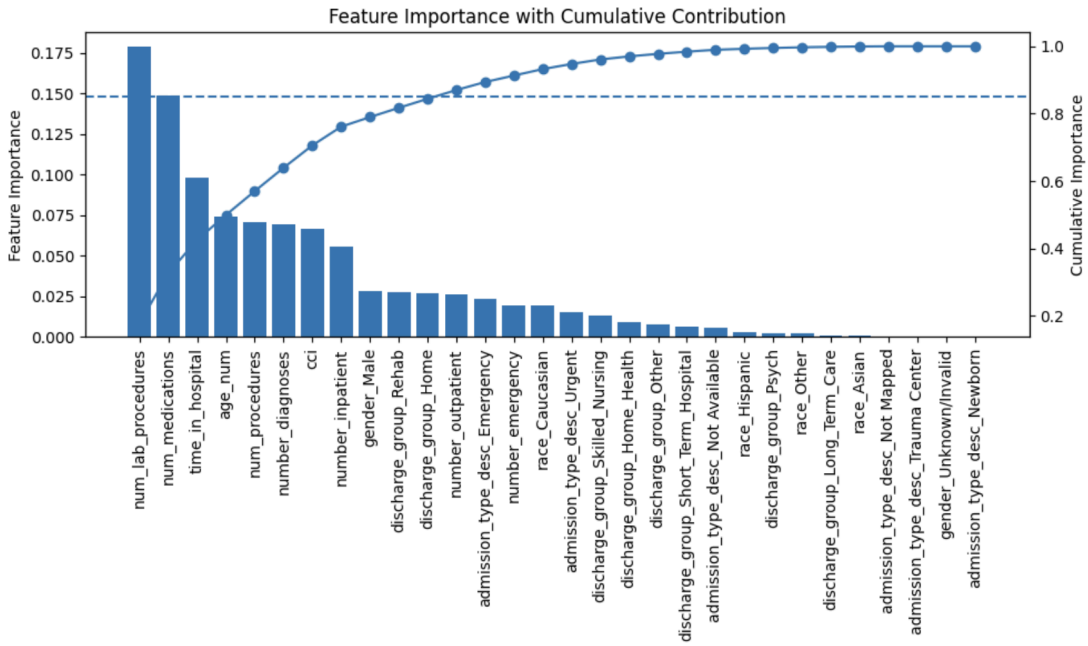

Figure 11. *Random Forest Feature Importance*

Given the unique properties of this ensemble method, specifically the incorporation of bootstrap sampling, random feature selection, and ensemble averaging to improve predictive performance, the Random Forest algorithm seemed promising for this classification problem; the results indicate that in this dataset, the model did not effectively distinguish between classes. This highlights that even with reduced variance and decorrelated trees, the ensemble may still struggle when the underlying data structure leads each bootstrap-sampled tree to favor one class over another.

Comparison of Models

Fold	Logistic Test AUC	Random Forest AUC
1	0.654702	0.647218
2	0.634180	0.636893
3	0.636361	0.634701
4	0.650684	0.649182
5	0.647454	0.648571
6	0.632268	0.625220
7	0.644578	0.647900

Mean Test AUC	0.6428894619414738	0.6413836034802465

Figure 12. *K-fold Cross Validation Comparison of Models' AUC*

A paired t-test conducted on cross-validated ROC AUC scores showed no statistically significant difference between the Random Forest and Logistic Regression models ($p = 0.397$). This suggests that any observed differences in performance are likely due to fold-level variability rather than a true improvement in predictive ability.

Conclusions

Prior research has shown that even experienced clinicians are often unable to predict readmissions more accurately than chance, because readmissions are influenced by numerous hard-to-measure factors and often rare compared to the whole patient population, resulting in class imbalance (Ebell et al., 1996).

This reality is evident in the results of both Logistic Regression and Random Forest models, which demonstrated moderate discriminatory performance with cross-validated ROC AUC values around 0.64. While the Random Forest model slightly outperformed Logistic Regression in some folds, a paired t-test confirmed that this difference was not statistically significant ($p = 0.397$). Despite similar AUC scores, both models struggled to accurately identify readmitted patients.

From an interpretability standpoint, both models identified consistent predictors of readmission risk. Variables related to prior healthcare utilization, such as number of inpatient visits, emergency visits, comorbidity burden (CCI), and discharge disposition, emerged as the most influential features. These findings align with domain knowledge and suggest that patients with greater clinical complexity and prior interactions with the system are at greater risk of readmission.

Overall, this analysis demonstrates that while machine learning models can provide insight into factors associated with readmissions, their predictive utility in this context remains limited. Nonetheless, the features identified in this study can serve as a starting point for decision-makers who want to explore further how these variables affect readmission rates. Because predictive performance is so poor for both a simple model like logistic regression and a model that usually reaches high predictive accuracy like random forests, one may lean towards more interpretable models like logistic regression to explore the impact of features further.

Sources

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Appendix

Audit of All Rows

	column	dtype	null_count	null_pct	variance	top_values	n_unique
weight	weight	object	68665	0.960108	NaN	{'[75-100)': 1195, '[50-75)': 781, '[100-125)'...	9
max_glu_serum	max_glu_serum	object	68062	0.951677	NaN	{'Norm': 1731, '>200': 969, '>300': 756}	3
A1Cresult	A1Cresult	object	58532	0.818423	NaN	{'>8': 6304, 'Norm': 3791, '>7': 2891}	3
medical_specialty	medical_specialty	object	34477	0.482074	NaN	{'InternalMedicine': 10919, 'Family/GeneralPra...	70
payer_code	payer_code	object	31043	0.434059	NaN	{'MC': 20414, 'HM': 4022, 'BC': 3432}	17
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admission_type_desc	admission_type_desc	object	4588	0.064152	NaN	{'Emergency': 36490, 'Elective': 13917, 'Urgen...	7
discharge_disposition_desc	discharge_disposition_desc	object	2474	0.034593	NaN	{'Discharged to home': 44317, 'Discharged/tran...	25
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metformin-rosiglitazone	metformin-rosiglitazone	object	0	0.000000	NaN	{'No': 71516, 'Steady': 2}	2
pioglitazone	pioglitazone	object	0	0.000000	NaN	{'No': 66210, 'Steady': 5047, 'Up': 180}	4
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diabetesMed	diabetesMed	object	0	0.000000	NaN	{'Yes': 54319, 'No': 17199}	2
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diag_1_cat	diag_1_cat	object	0	0.000000	NaN	{'Circulatory': 21821, 'Endocrine': 7685, 'Res...	20
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rosiglitazone	rosiglitazone	object	0	0.000000	NaN	{'No': 66817, 'Steady': 4490, 'Up': 136}	4
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num_lab_procedures	num_lab_procedures	int64	0	0.000000	3.980958e+02	None	116
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nateglinide	nateglinide	object	0	0.000000	NaN	{'No': 71020, 'Steady': 474, 'Up': 16}	4
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glimepiride	glimepiride	object	0	0.000000	NaN	{'No': 67773, 'Steady': 3372, 'Up': 235}	4
patient_nbr	patient_nbr	int64	0	0.000000	1.558333e+15	None	71518
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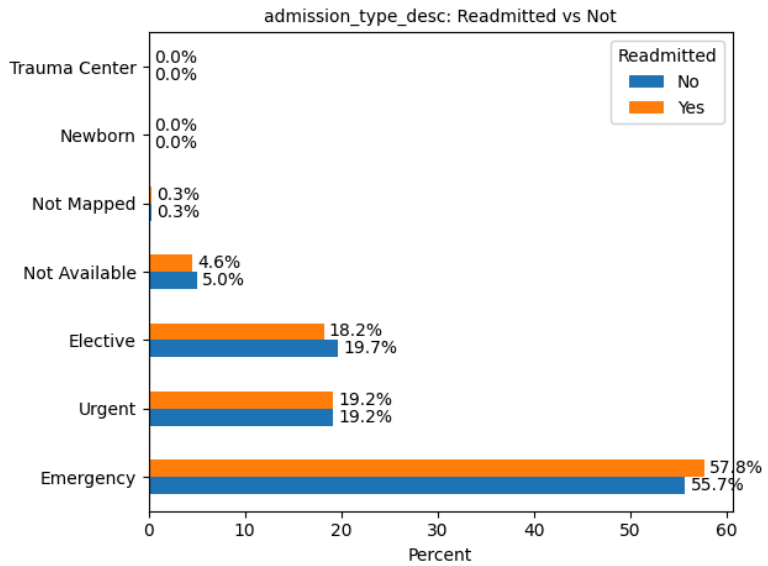
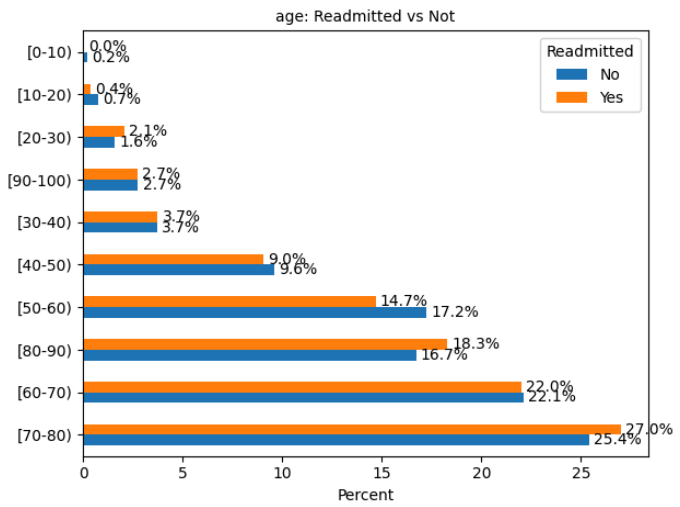
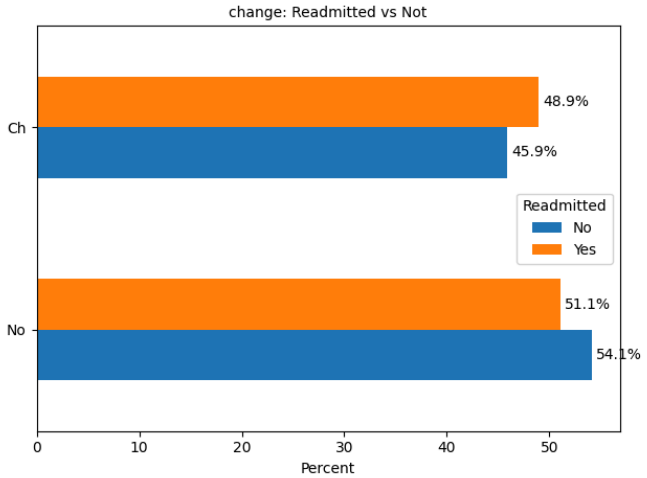
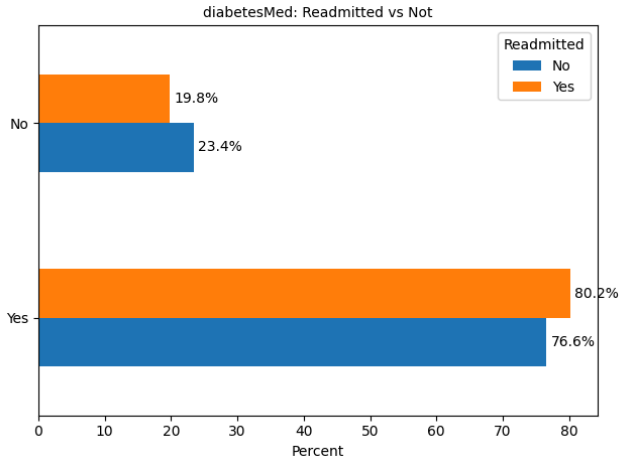
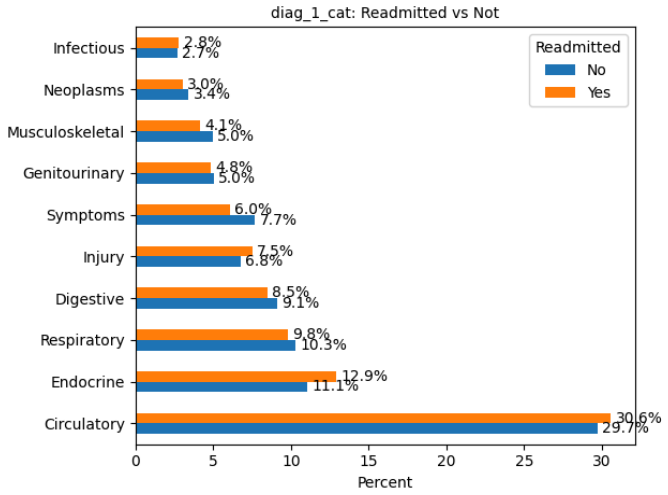
Frequency Plots of Categorical Vars

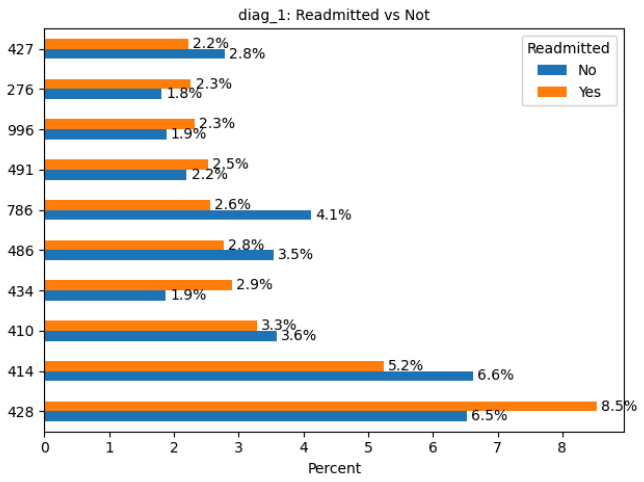


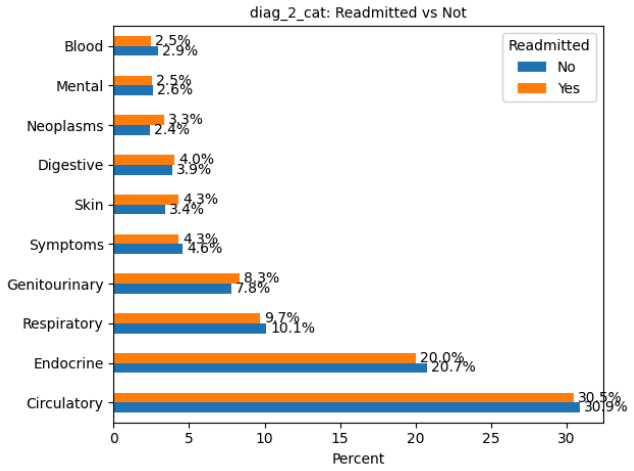


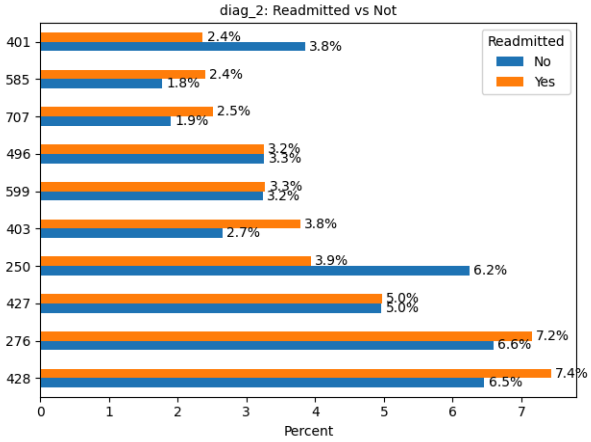

change: Readmitted vs Not
Ch
No
48.9%
45.9%
51.1%
54.1%
Readmitted
No
Yes
0
10
20
30
40
50
Percent

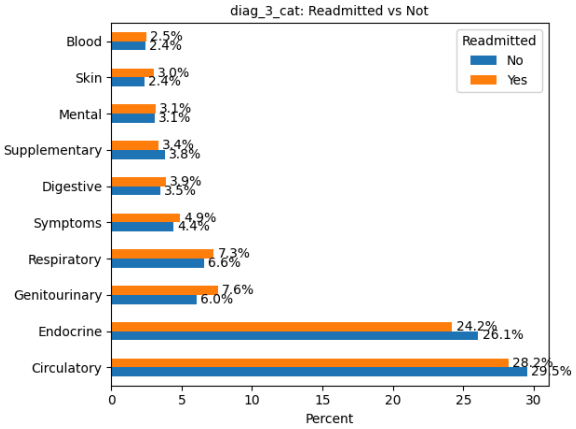

diabetesMed: Readmitted vs Not
No
Yes
19.8%
23.4%
80.2%
76.6%
Readmitted
No
Yes
0
10
20
30
40
50
60
70
80
Percent

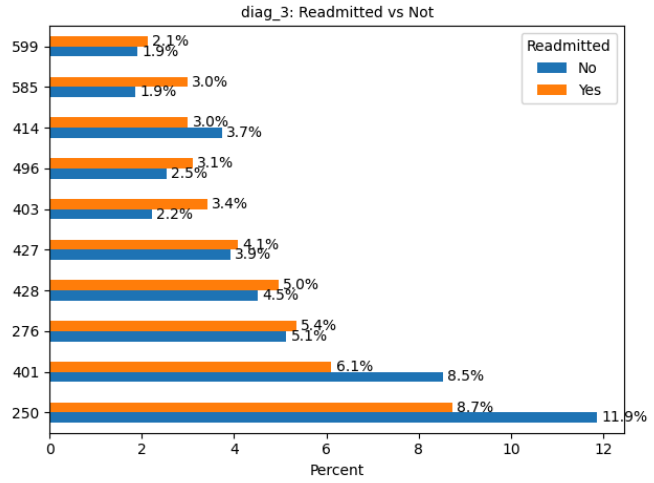

diag_1_cat: Readmitted vs Not
Infectious
Neoplasms
Musculoskeletal
Genitourinary
Symptoms
Injury
Digestive
Respiratory
Endocrine
Circulatory
2.8%
2.7%
3.0%
3.4%
4.1%
5.0%
4.8%
5.0%
6.0%
7.7%
7.5%
6.8%
8.5%
9.1%
9.8%
10.3%
12.9%
11.1%
30.6%
29.7%
Readmitted
No
Yes
0
5
10
15
20
25
30
Percent

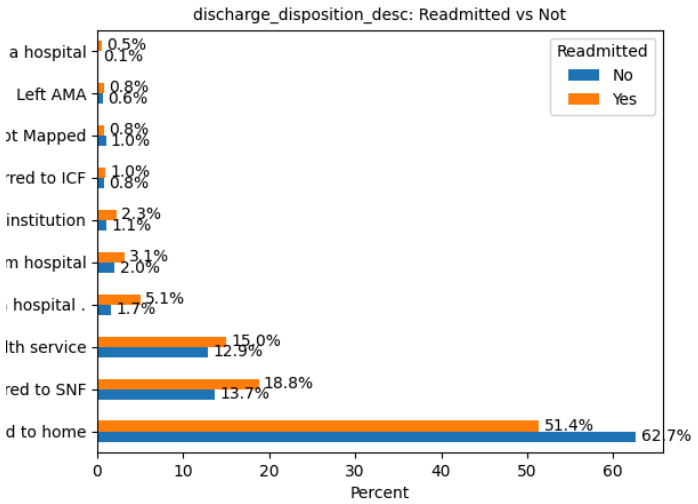

diag_1: Readmitted vs Not
Readmitted
No
Yes
427
276
996
491
786
486
434
410
414
428
2.2%
2.8%
2.3%
1.8%
2.3%
1.9%
2.5%
2.2%
2.6%
4.1%
2.8%
3.5%
2.9%
1.9%
3.3%
3.6%
5.2%
6.6%
8.5%
6.5%
0
1
2
3
4
5
6
7
8
Percent

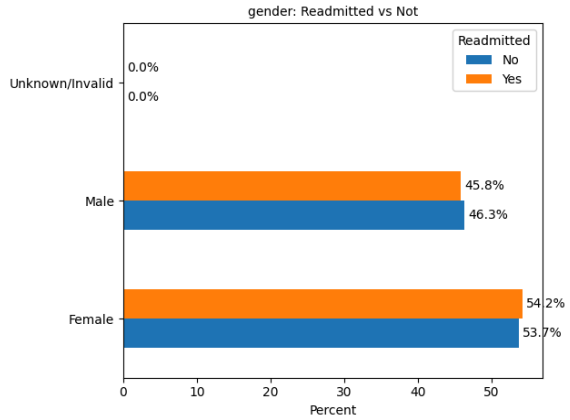

diag_2_cat: Readmitted vs Not
Readmitted
No
Yes
Blood
Mental
Neoplasms
Digestive
Skin
Symptoms
Genitourinary
Respiratory
Endocrine
Circulatory
2.5%
2.9%
2.5%
2.6%
3.3%
2.4%
4.0%
3.9%
4.3%
3.4%
4.3%
4.6%
8.3%
7.8%
9.7%
10.1%
20.0%
20.7%
30.5%
30.9%
0
5
10
15
20
25
30
Percent

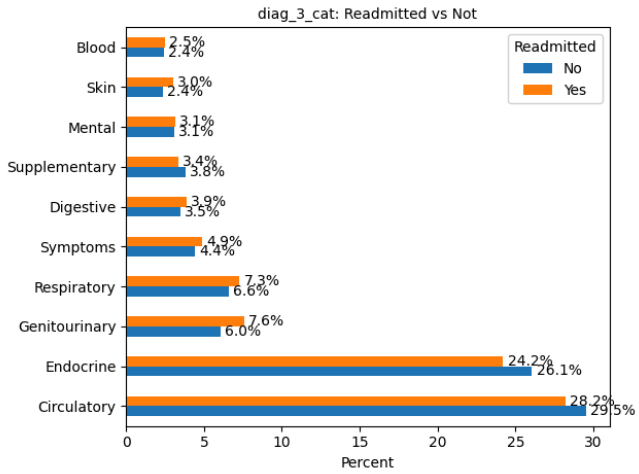

diag_2: Readmitted vs Not
Readmitted
No
Yes
401
585
707
496
599
403
250
427
276
428
2.4%
3.8%
2.4%
1.8%
2.5%
1.9%
3.2%
3.3%
3.3%
3.2%
3.8%
2.7%
3.9%
6.2%
5.0%
5.0%
7.2%
6.6%
7.4%
6.5%
0
1
2
3
4
5
6
7
Percent

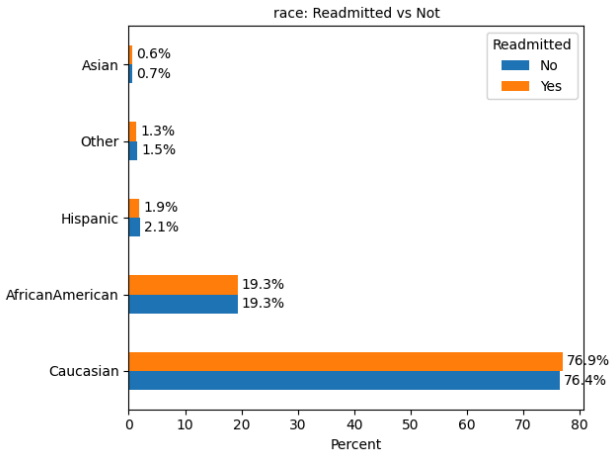

diag_3_cat: Readmitted vs Not
Readmitted
No
Yes
Blood
Skin
Mental
Supplementary
Digestive
Symptoms
Respiratory
Genitourinary
Endocrine
Circulatory
2.5%
2.4%
3.0%
2.4%
3.1%
3.1%
3.4%
3.8%
3.9%
3.5%
4.9%
4.4%
7.3%
6.6%
7.6%
6.0%
24.2%
26.1%
28.2%
29.5%
0
5
10
15
20
25
30
Percent


diag_3: Readmitted vs Not
Readmitted
No
Yes
599
585
414
496
403
427
428
276
401
250
2.1%
1.9%
3.0%
1.9%
3.0%
3.7%
3.1%
2.5%
3.4%
2.2%
4.1%
3.9%
5.0%
4.5%
5.4%
5.1%
6.1%
8.5%
8.7%
11.9%
0
2
4
6
8
10
12
Percent


discharge_disposition_desc: Readmitted vs Not
Readmitted
No
Yes
a hospital
Left AMA
ot Mapped
red to ICF
institution
m hospital
hospital .
lth service
red to SNF
d to home
0.5%
0.1%
0.8%
0.6%
0.8%
1.0%
1.0%
0.8%
2.3%
1.1%
3.1%
2.0%
5.1%
1.7%
15.0%
12.9%
18.8%
13.7%
51.4%
62.7%
0
10
20
30
40
50
60
Percent


gender: Readmitted vs Not
Readmitted
No
Yes
Unknown/Invalid
Male
Female
0.0%
0.0%
45.8%
46.3%
54.2%
53.7%
0
10
20
30
40
50
Percent





Discharge to Hospice or Death by CCI

```
cci
0     0.022548
1     0.026636
2     0.018165
3     0.024218
4     0.025146
5     0.044715
6     0.135135
7     0.100000
8     0.147760
9     0.239130
10    0.072000
11    0.500000
```

Logit Regression Results

```
Warning: Maximum number of iterations has been exceeded.
         Current function value: 0.290316
         Iterations: 35
                           Logit Regression Results
==============================================================================
Dep. Variable:         readmitted_<30   No. Observations:                55978
Model:                          Logit   Df Residuals:                    55947
Method:                           MLE   Df Model:                           30
Date:                Sun, 19 Apr 2026   Pseudo R-squ.:                 0.03829
Time:                        07:13:45   Log-Likelihood:                -16251.
converged:                       False   LL-Null:                       -16898.
Covariance Type:            nonrobust   LLR p-value:                2.568e-253
==============================================================================
========================
                                          coef    std err          z      P>|z|
[0.025      0.975]
------------------------------------------------------------------------------
------------------------
const                                   -3.2861      0.218    -15.100      0.000
-3.713      -2.860
time_in_hospital                         0.0211      0.006      3.555      0.000
0.009       0.033
num_lab_procedures                       0.0013      0.001      1.549      0.121
-0.000       0.003
num_procedures                          -0.0023      0.010     -0.238      0.812
-0.022       0.017
num_medications                          0.0022      0.002      0.979      0.327
-0.002       0.007
number_outpatient                        0.0001      0.014      0.008      0.994
-0.027       0.027
number_emergency                         0.1059      0.024      4.462      0.000
0.059       0.152
number_inpatient                         0.3464      0.018     19.144      0.000
0.311       0.382
number_diagnoses                         0.0379      0.009      4.329      0.000
0.021       0.055
cci                                      0.0768      0.010      7.586      0.000
0.057       0.097
gender_Male                              0.0273      0.030      0.897      0.370
-0.032       0.087
gender_Unknown/Invalid                 -16.9321   1.51e+04     -0.001      0.999
-2.95e+04    2.95e+04
race_Asian                              -0.1156      0.195     -0.592      0.554
-0.498       0.267
race_Caucasian                           0.0036      0.039      0.093      0.926
-0.072       0.080
```

```
race_Hispanic                          -0.0649      0.115     -0.566      0.571
-0.289       0.160
race_Other                             -0.1174      0.132     -0.891      0.373
-0.376       0.141
admission_type_desc_Emergency           0.0454      0.039      1.151      0.250
-0.032       0.123
admission_type_desc_Newborn           -12.8416    933.779     -0.014      0.989
-1843.015    1817.332
admission_type_desc_Not Available       0.0463      0.080      0.576      0.565
-0.111       0.204
admission_type_desc_Not Mapped         -0.0815      0.265     -0.307      0.759
-0.601       0.438
admission_type_desc_Trauma Center     -12.5516    361.392     -0.035      0.972
-720.867     695.764
admission_type_desc_Urgent              0.0356      0.047      0.751      0.453
-0.057       0.128
discharge_group_Home                   -0.2747      0.199     -1.377      0.168
-0.666       0.116
discharge_group_Home_Health            -0.0491      0.204     -0.241      0.810
-0.448       0.350
discharge_group_Long_Term_Care         -1.7601      0.544     -3.237      0.001
-2.826      -0.694
discharge_group_Other                   0.3419      0.206      1.659      0.097
-0.062       0.746
discharge_group_Psych                   1.5005      0.330      4.547      0.000
0.854       2.147
discharge_group_Rehab                   1.2088      0.211      5.718      0.000
0.794       1.623
discharge_group_Short_Term_Hospital     0.4367      0.215      2.032      0.042
0.015       0.858
discharge_group_Skilled_Nursing         0.2957      0.203      1.456      0.145
-0.102       0.694
age_num                                 0.0048      0.001      4.434      0.000
0.003       0.007
==============================================================================
========================

Significant Variables:
['time_in_hospital', 'number_emergency', 'number_inpatient',
'number_diagnoses', 'cci', 'discharge_group_Long_Term_Care',
'discharge_group_Psych', 'discharge_group_Rehab',
'discharge_group_Short_Term_Hospital', 'age_num']
Optimization terminated successfully.
         Current function value: 0.292402
         Iterations 8
```

Random Forest Feature Importance

```
 Random Forest (Tuned)

Accuracy: 0.91

F1 Score: 0.0

AUC: 0.637

Confusion Matrix:

[[12740     0]

 [ 1255     0]]

Feature Importance:

                                Feature    Importance

1                    num_lab_procedures  1.785846e-01

3                       num_medications  1.486213e-01

0                      time_in_hospital  9.839547e-02

29                             age_num  7.410257e-02

2                        num_procedures  7.068947e-02

7                      number_diagnoses  6.920682e-02

8                                   cci  6.648489e-02

6                      number_inpatient  5.548637e-02

9                          gender_Male  2.844374e-02

26                discharge_group_Rehab  2.747802e-02

21                 discharge_group_Home  2.668131e-02

4                      number_outpatient  2.615412e-02
```

15	admission_type_desc_Emergency	2.375765e-02
5	number_emergency	1.921749e-02
12	race_Caucasian	1.914987e-02
20	admission_type_desc_Urgent	1.513732e-02
28	discharge_group_Skilled_Nursing	1.343969e-02
22	discharge_group_Home_Health	9.058019e-03
24	discharge_group_Other	7.530153e-03
27	discharge_group_Short_Term_Hospital	6.458911e-03
17	admission_type_desc_Not Available	5.621545e-03
13	race_Hispanic	2.930695e-03
25	discharge_group_Psych	2.485047e-03
14	race_Other	2.008910e-03
23	discharge_group_Long_Term_Care	1.238746e-03
11	race_Asian	1.087711e-03
18	admission_type_desc_Not Mapped	5.490219e-04
19	admission_type_desc_Trauma Center	5.050059e-07
10	gender_Unknown/Invalid	0.000000e+00